Drill Bit Size Chart

Fractional and letter sizes — 1/64 in to 1 in, A to Z

90 sizes
fractional + letter
sheet 1 of 3

1/64 – 1/2 — 32 sizes

FRACTION	INCH	MM
1/64	0.0156	0.397
1/32	0.0313	0.794
3/64	0.0469	1.191
1/16	0.0625	1.588
5/64	0.0781	1.984
3/32	0.0938	2.381
7/64	0.1094	2.778
1/8	0.1250	3.175
9/64	0.1406	3.572
5/32	0.1563	3.969
11/64	0.1719	4.366
3/16	0.1875	4.763
13/64	0.2031	5.159
7/32	0.2188	5.556
15/64	0.2344	5.953
1/4	0.2500	6.350
17/64	0.2656	6.747
9/32	0.2813	7.144
19/64	0.2969	7.541
5/16	0.3125	7.938
21/64	0.3281	8.334
11/32	0.3438	8.731
23/64	0.3594	9.128
3/8	0.3750	9.525
25/64	0.3906	9.922
13/32	0.4063	10.319
27/64	0.4219	10.716
7/16	0.4375	11.113
29/64	0.4531	11.509
15/32	0.4688	11.906
31/64	0.4844	12.303
1/2	0.5000	12.700

33/64 – 1 — 32 sizes

FRACTION	INCH	MM
33/64	0.5156	13.097
17/32	0.5313	13.494
35/64	0.5469	13.891
9/16	0.5625	14.288
37/64	0.5781	14.684
19/32	0.5938	15.081
39/64	0.6094	15.478
5/8	0.6250	15.875
41/64	0.6406	16.272
21/32	0.6563	16.669
43/64	0.6719	17.066
11/16	0.6875	17.463
45/64	0.7031	17.859
23/32	0.7188	18.256
47/64	0.7344	18.653
3/4	0.7500	19.050
49/64	0.7656	19.447
25/32	0.7813	19.844
51/64	0.7969	20.241
13/16	0.8125	20.638
53/64	0.8281	21.034
27/32	0.8438	21.431
55/64	0.8594	21.828
7/8	0.8750	22.225
57/64	0.8906	22.622
29/32	0.9063	23.019
59/64	0.9219	23.416
15/16	0.9375	23.813
61/64	0.9531	24.209
31/32	0.9688	24.606
63/64	0.9844	25.003
1	1.0000	25.400

A – Z — 26 sizes

LETTER	INCH	MM
A	0.2340	5.944
B	0.2380	6.045
C	0.2420	6.147
D	0.2460	6.248
E	0.2500	6.350
F	0.2570	6.528
G	0.2610	6.629
H	0.2660	6.756
I	0.2720	6.909
J	0.2770	7.036
K	0.2810	7.137
L	0.2900	7.366
M	0.2950	7.493
N	0.3020	7.671
O	0.3160	8.026
P	0.3230	8.204
Q	0.3320	8.433
R	0.3390	8.611
S	0.3480	8.839
T	0.3580	9.093
U	0.3680	9.347
V	0.3770	9.576
W	0.3860	9.804
X	0.3970	10.084
Y	0.4040	10.262
Z	0.4130	10.490

THE SAME DRILL HAS TWO PUBLISHED INCH VALUES

A 64th is an exact fraction, but the printed decimal is a choice. Charts that round half-up and charts that truncate disagree on 32 of the 64 fractional sizes — every odd numerator, because n/64 always lands on a fifth decimal place of 5:

1/32 → **0.0313** rounded, 0.0312 truncated
5/32 → **0.1563** rounded, 0.1562 truncated
19/64 → **0.2969** rounded, 0.2968 truncated
11/32 → **0.3438** rounded, 0.3437 truncated

These sheets round half-up throughout. The truncated figures are not wrong measurements, they are the same exact value cut short.

WHERE THESE NUMBERS COME FROM

Fractional sizes — n/64 in, exact. 1/64 to 1 in is the whole series; 1/2 in and larger is also sold as reduced-shank.

inch column — the exact value to four places, half-up.

mm column — exact value × 25.4, three places — **not** the four-place inch figure × 25.4. That shortcut is where the other published mm value comes from: 3/32 is 2.381 this way and 2.383 from the rounded inch.

Letter sizes — ANSI/ASME B94.11M, A 0.2340 in to Z 0.4130 in, 26 sizes. K alone is printed as 0.2811 by one major chart; two others and the standard give **0.2810**, so 0.2810 is what appears here.

Which to trust — the fraction. It is exact by construction; both printed decimals are conventions applied to it.

Drill Bit Size Chart

Number drills — #1 to #80

#1 – #40 — 40 sizes

NUMBER	INCH	MM
#1	0.2280	5.791
#2	0.2210	5.613
#3	0.2130	5.410
#4	0.2090	5.309
#5	0.2055	5.220
#6	0.2040	5.182
#7	0.2010	5.105
#8	0.1990	5.055
#9	0.1960	4.978
#10	0.1935	4.915
#11	0.1910	4.851
#12	0.1890	4.801
#13	0.1850	4.699
#14	0.1820	4.623
#15	0.1800	4.572
#16	0.1770	4.496
#17	0.1730	4.394
#18	0.1695	4.305
#19	0.1660	4.216
#20	0.1610	4.089
#21	0.1590	4.039
#22	0.1570	3.988
#23	0.1540	3.912
#24	0.1520	3.861
#25	0.1495	3.797
#26	0.1470	3.734
#27	0.1440	3.658
#28	0.1405	3.569
#29	0.1360	3.454
#30	0.1285	3.264
#31	0.1200	3.048
#32	0.1160	2.946
#33	0.1130	2.870
#34	0.1110	2.819
#35	0.1100	2.794
#36	0.1065	2.705
#37	0.1040	2.642
#38	0.1015	2.578
#39	0.0995	2.527
#40	0.0980	2.489

#41 – #80 — 40 sizes

NUMBER	INCH	MM
#41	0.0960	2.438
#42	0.0935	2.375
#43	0.0890	2.261
#44	0.0860	2.184
#45	0.0820	2.083
#46	0.0810	2.057
#47	0.0785	1.994
#48	0.0760	1.930
#49	0.0730	1.854
#50	0.0700	1.778
#51	0.0670	1.702
#52	0.0635	1.613
#53	0.0595	1.511
#54	0.0550	1.397
#55	0.0520	1.321
#56	0.0465	1.181
#57	0.0430	1.092
#58	0.0420	1.067
#59	0.0410	1.041
#60	0.0400	1.016
#61	0.0390	0.991
#62	0.0380	0.965
#63	0.0370	0.940
#64	0.0360	0.914
#65	0.0350	0.889
#66	0.0330	0.838
#67	0.0320	0.813
#68	0.0310	0.787
#69	0.0292	0.742
#70	0.0280	0.711
#71	0.0260	0.660
#72	0.0250	0.635
#73	0.0240	0.610
#74	0.0225	0.572
#75	0.0210	0.533
#76	0.0200	0.508
#77	0.0180	0.457
#78	0.0160	0.406
#79	0.0145	0.368
#80	0.0135	0.343

NUMBER DRILLS COUNT BACKWARDS

#1 is the largest at 0.2280 in and #80 the smallest at 0.0135 in. The number is a catalogue index, not a dimension — there is no arithmetic that turns #37 into a diameter. The steps are not even either: the series tightens as it falls, so the first twenty sizes are spread over 0.0670 in while the last twenty fit inside 0.0255 in.

#1 → **0.2280 in** · #80 → **0.0135 in**
#1–#20 0.0670 in over 20 sizes
#61–#80 0.0255 in over 20 sizes

Below #60 the whole range fits inside 0.0265 in, so a chart lets you pick the next size up without guessing at the print.

WHERE THESE NUMBERS COME FROM

Definition — ANSI/ASME B94.11M fixed inch values, not a formula. Every figure in the inch column is the standard value printed to four places.

mm column — standard value × 25.4, three places, half-up.

Cross-check — all 80 values agree across three independent published charts. The number series is the least contested of the four on this sheet.

Use — number drills are for tap and clearance holes. The tap drill for a given thread is a number size, which is why this column exists next to the fractional one.

Not interchangeable — #30 is 0.1285 in and 1/8 in is 0.1250 in. Close, but a clearance hole cut with the wrong one is either loose or undersized.

Drill Bit Size Chart

Metric drills — ISO 235 preferred sizes, 1.0 to 25.0 mm

1.0 – 4.0 MM — 31 sizes

METRIC	INCH
1.0	0.0394
1.1	0.0433
1.2	0.0472
1.3	0.0512
1.4	0.0551
1.5	0.0591
1.6	0.0630
1.7	0.0669
1.8	0.0709
1.9	0.0748
2.0	0.0787
2.1	0.0827
2.2	0.0866
2.3	0.0906
2.4	0.0945
2.5	0.0984
2.6	0.1024
2.7	0.1063
2.8	0.1102
2.9	0.1142
3.0	0.1181
3.1	0.1220
3.2	0.1260
3.3	0.1299
3.4	0.1339
3.5	0.1378
3.6	0.1417
3.7	0.1457
3.8	0.1496
3.9	0.1535
4.0	0.1575

4.1 – 7.0 MM — 30 sizes

METRIC	INCH
4.1	0.1614
4.2	0.1654
4.3	0.1693
4.4	0.1732
4.5	0.1772
4.6	0.1811
4.7	0.1850
4.8	0.1890
4.9	0.1929
5.0	0.1969
5.1	0.2008
5.2	0.2047
5.3	0.2087
5.4	0.2126
5.5	0.2165
5.6	0.2205
5.7	0.2244
5.8	0.2283
5.9	0.2323
6.0	0.2362
6.1	0.2402
6.2	0.2441
6.3	0.2480
6.4	0.2520
6.5	0.2559
6.6	0.2598
6.7	0.2638
6.8	0.2677
6.9	0.2717
7.0	0.2756

7.1 – 10.0 MM — 30 sizes

METRIC	INCH
7.1	0.2795
7.2	0.2835
7.3	0.2874
7.4	0.2913
7.5	0.2953
7.6	0.2992
7.7	0.3031
7.8	0.3071
7.9	0.3110
8.0	0.3150
8.1	0.3189
8.2	0.3228
8.3	0.3268
8.4	0.3307
8.5	0.3346
8.6	0.3386
8.7	0.3425
8.8	0.3465
8.9	0.3504
9.0	0.3543
9.1	0.3583
9.2	0.3622
9.3	0.3661
9.4	0.3701
9.5	0.3740
9.6	0.3780
9.7	0.3819
9.8	0.3858
9.9	0.3898
10.0	0.3937

10.5 – 25.0 MM — 30 sizes

METRIC	INCH
10.5	0.4134
11.0	0.4331
11.5	0.4528
12.0	0.4724
12.5	0.4921
13.0	0.5118
13.5	0.5315
14.0	0.5512
14.5	0.5709
15.0	0.5906
15.5	0.6102
16.0	0.6299
16.5	0.6496
17.0	0.6693
17.5	0.6890
18.0	0.7087
18.5	0.7283
19.0	0.7480
19.5	0.7677
20.0	0.7874
20.5	0.8071
21.0	0.8268
21.5	0.8465
22.0	0.8661
22.5	0.8858
23.0	0.9055
23.5	0.9252
24.0	0.9449
24.5	0.9646
25.0	0.9843

METRIC SETS CHANGE STEP AT 10 MM

From 1.0 to 10.0 mm the series steps by 0.1 mm, so a set has 91 sizes in that band alone. Above 10 mm it jumps to 0.5 mm steps and 30 more sizes reach 25.0 mm. That is why a metric drill index looks crowded at the bottom and sparse at the top.

1.0 → **0.3937 in** · 25.0 → **0.9843 in**
1.0–10.0 mm 0.1 mm steps 91 sizes
10.5–25.0 mm 0.5 mm steps 30 sizes

Converting the other way — an inch drill to metric — will almost never land on a stocked size. Buy the nearest metric drill, not the exact conversion.

WHERE THESE NUMBERS COME FROM

Definition — ISO 235 preferred sizes, millimetres nominal. The mm column is the definition; the inch column is derived.

inch column — mm ÷ 25.4, four places, half-up. 25.4 is exact by the 1959 agreement, so the only rounding is the four-place cut.

Reading the sheet — sizes are listed by millimetre, which is how they are stamped and sold. Column order is deliberate: the definition on the left, the conversion on the right, so a typo in the conversion cannot pass for a definition.

Coverage — 1.0 to 25.0 mm is the hand-drill and drill-press range. Above 25 mm most shops move to a taper-shank or an annular cutter.